

TITLE:
IAPT Referral-to-Treatment (RTT) Analysis

Dataset Used:
nhstalkingtherapies_month_mar_2026_activity_performance.csv

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IAPT Referral-to-Treatment (RTT) Analysis

Introduction

This project analyses **Improving Access to Psychological Therapies (IAPT)** referral-to-treatment (RTT) waiting times using an IAPT dataset. The purpose of the analysis is to measure how long patients wait between their referral date and the start of treatment.

The analysis calculates:

- **Mean RTT waiting time**
- **Median RTT waiting time**

across:

- Different **regions**
- Different **mental health disorder types**

This helps identify:

- Regional differences in service delivery
- Treatment accessibility trends
- Disorders associated with longer waiting periods
- Potential NHS resource and capacity challenges

The screenshot shows a Google Colab notebook titled "week 3 ana 1 (aiswarya)". The code in the notebook is as follows:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

file_path = "/content/nhsstalkingtherapies_month_mar_2026_activity_performance.csv"

# Read dataset
_df = pd.read_csv(file_path)

# Display first 5 rows
_df.head()
```

The output of the code shows the first 5 rows of the dataset:

	REPORTING_PERIOD_START	REPORTING_PERIOD_END	GROUP_TYPE	ORG_CODE1	ORG_NAME1	ORG_CODE2	ORG_NAME2	MEASURE_ID	MEASURE_NAME	MEASURE_VALUE_SUPPRESSED
0	2026-03-01	2026-03-31	England	all	all SubICBs	all	all Providers	M001	Count_ReferralsReceived	163100
1	2026-03-01	2026-03-31	England	all	all SubICBs	all	all Providers	M002	Count_SelfReferrals	111630
2	2026-03-01	2026-03-31	England	all	all SubICBs	all	all Providers	M003	Count_GPRReferrals	13461
3	2026-03-01	2026-03-31	England	all	all SubICBs	all	all Providers	M004	Count_HealthVisitorReferrals	433
4	2026-03-01	2026-03-31	England	all	all SubICBs	all	all Providers	M005	Count_OtherPrimaryCareReferrals	5840

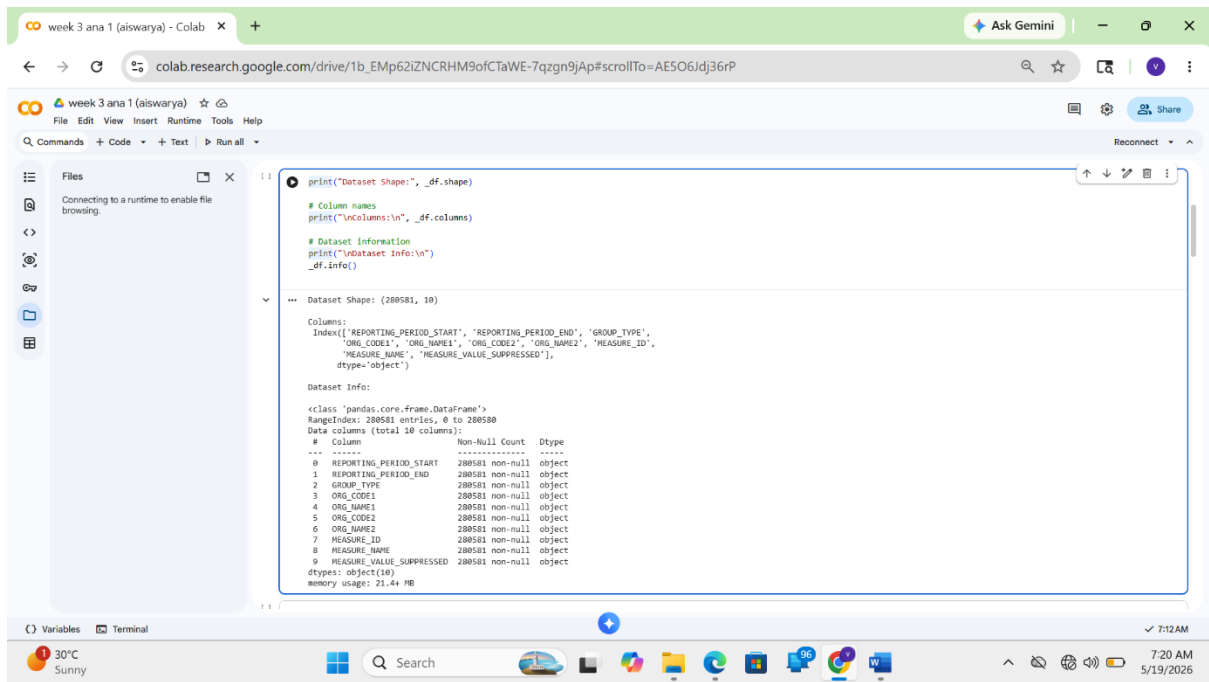
The notebook also shows the dataset shape and column names:

```
print("Dataset Shape:", _df.shape)

# Column names
print("\nColumns:\n", _df.columns)

# Dataset information
print("\nDataset Info:\n")
_df.info()
```

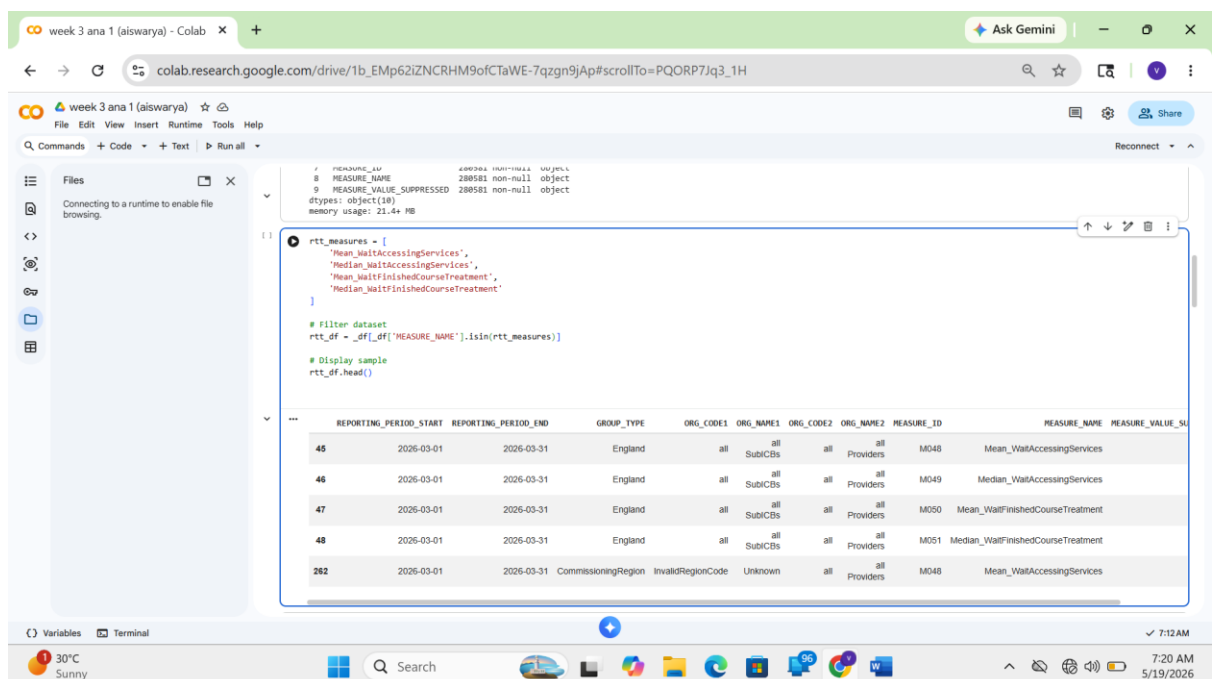
The bottom of the notebook shows the Windows taskbar with the date 5/19/2026 and time 7:12 AM.



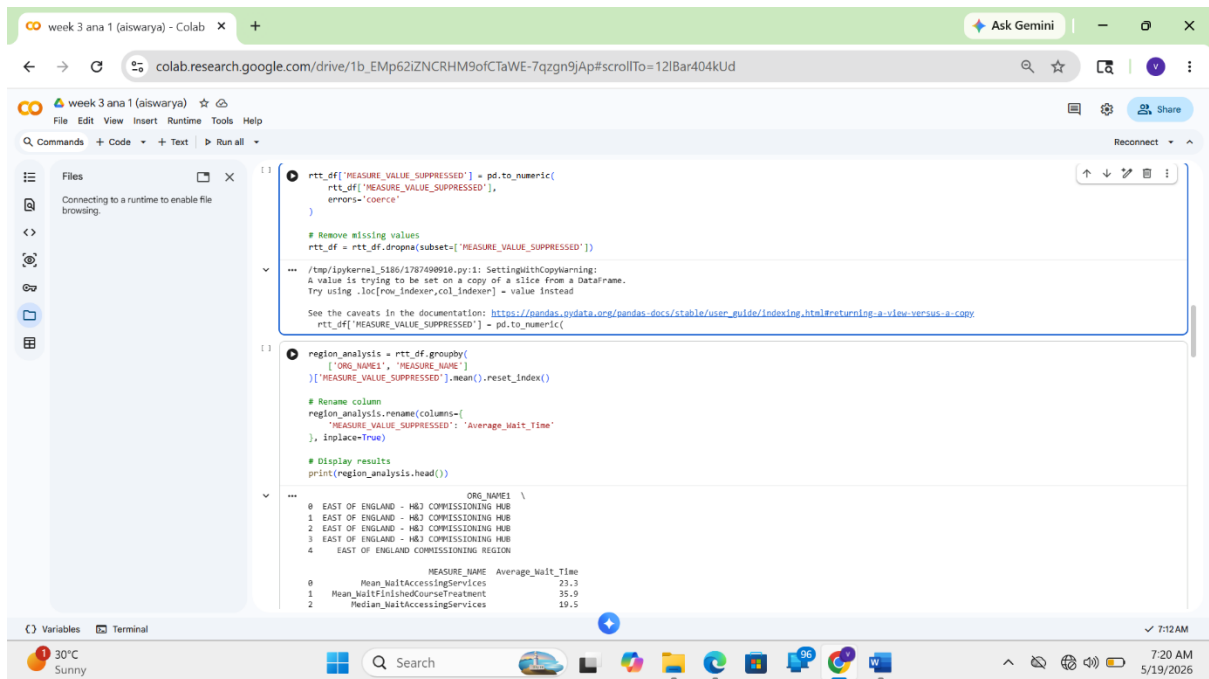
RTT by Region Table

This table shows:

- Average waiting time in each region
- Median waiting time for more stable comparison
- Regional variation in access to treatment



Convert Measure Values to Numeric



The screenshot shows a Google Colab notebook with the following code and output:

```
rtt_df["MEASURE_VALUE_SUPPRESSED"] = pd.to_numeric(
    rtt_df["MEASURE_VALUE_SUPPRESSED"],
    errors="coerce"
)

# Remove missing values
rtt_df = rtt_df.dropna(subset=["MEASURE_VALUE_SUPPRESSED"])

region_analysis = rtt_df.groupby(
    ["ORG_NAME1", "MEASURE_NAME"]
)[["MEASURE_VALUE_SUPPRESSED"]].mean().reset_index()

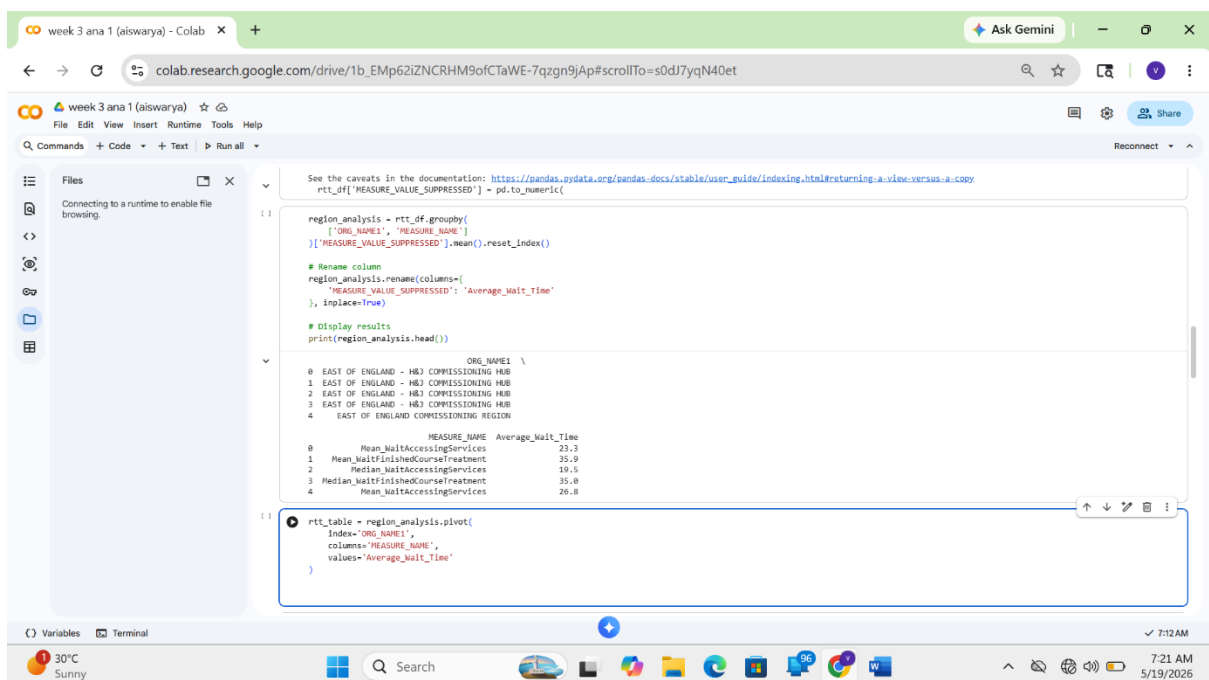
# Rename column
region_analysis.rename(columns={
    "MEASURE_VALUE_SUPPRESSED": "Average_Wait_Time"
}, inplace=True)

# Display results
print(region_analysis.head())
```

The output shows a table with 5 rows and 3 columns:

	ORG_NAME1 \	MEASURE_NAME	Average_Wait_Time
0	EAST OF ENGLAND - H&J COMMISSIONING HUB	Mean_WaitAccessingServices	23.3
1	EAST OF ENGLAND - H&J COMMISSIONING HUB	Mean_WaitFinishedCourseTreatment	35.9
2	EAST OF ENGLAND - H&J COMMISSIONING HUB	Median_WaitAccessingServices	19.5
3	EAST OF ENGLAND - H&J COMMISSIONING HUB		
4	EAST OF ENGLAND COMMISSIONING REGION		

Mean and Median RTT by Region



The screenshot shows a Google Colab notebook with the following code and output:

```
region_analysis = rtt_df.groupby(
    ["ORG_NAME1", "MEASURE_NAME"]
)[["MEASURE_VALUE_SUPPRESSED"]].mean().reset_index()

# Rename column
region_analysis.rename(columns={
    "MEASURE_VALUE_SUPPRESSED": "Average_Wait_Time"
}, inplace=True)

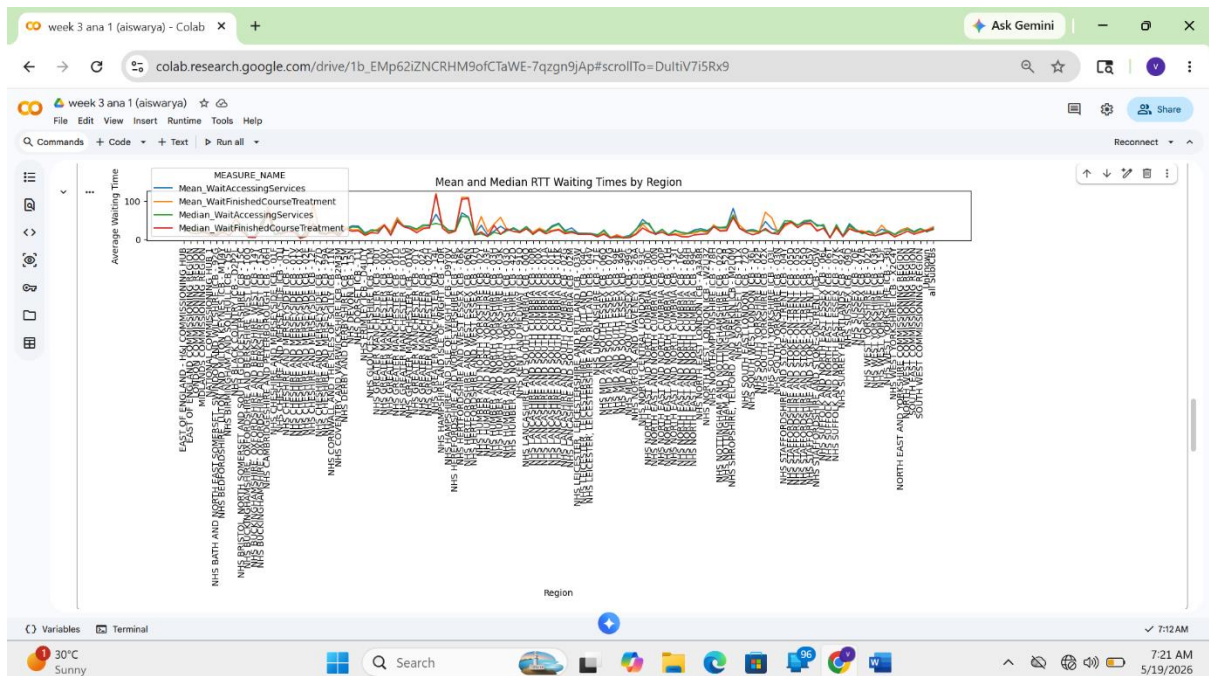
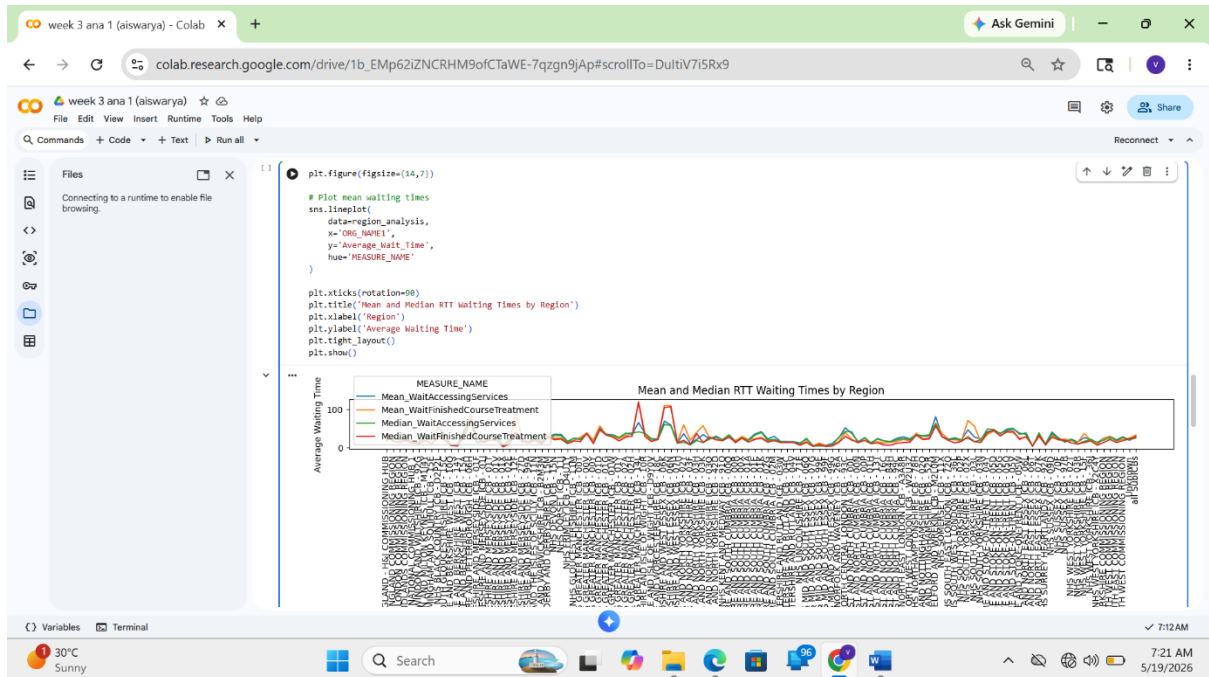
# Display results
print(region_analysis.head())

rtt_table = region_analysis.pivot(
    index="ORG_NAME1",
    columns="MEASURE_NAME",
    values="Average_Wait_Time"
)
```

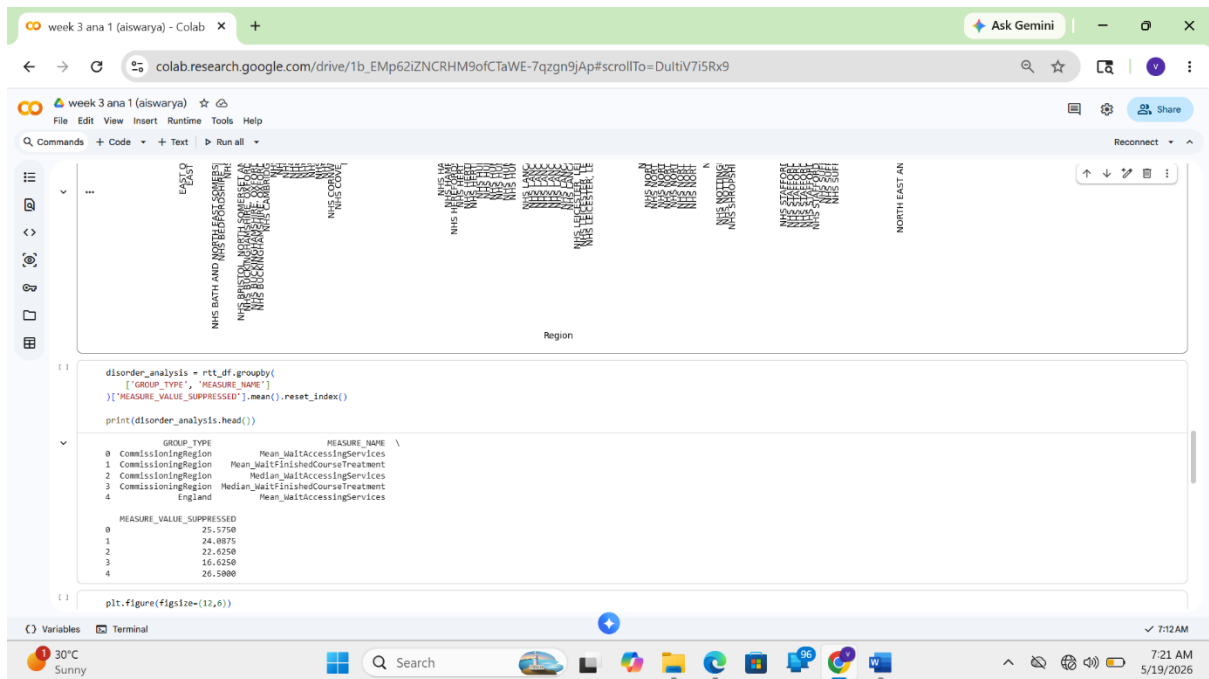
The output shows a table with 5 rows and 3 columns:

	ORG_NAME1 \	MEASURE_NAME	Average_Wait_Time
0	EAST OF ENGLAND - H&J COMMISSIONING HUB	Mean_WaitAccessingServices	23.3
1	EAST OF ENGLAND - H&J COMMISSIONING HUB	Mean_WaitFinishedCourseTreatment	35.9
2	EAST OF ENGLAND - H&J COMMISSIONING HUB	Median_WaitAccessingServices	19.5
3	EAST OF ENGLAND - H&J COMMISSIONING HUB		
4	EAST OF ENGLAND COMMISSIONING REGION		

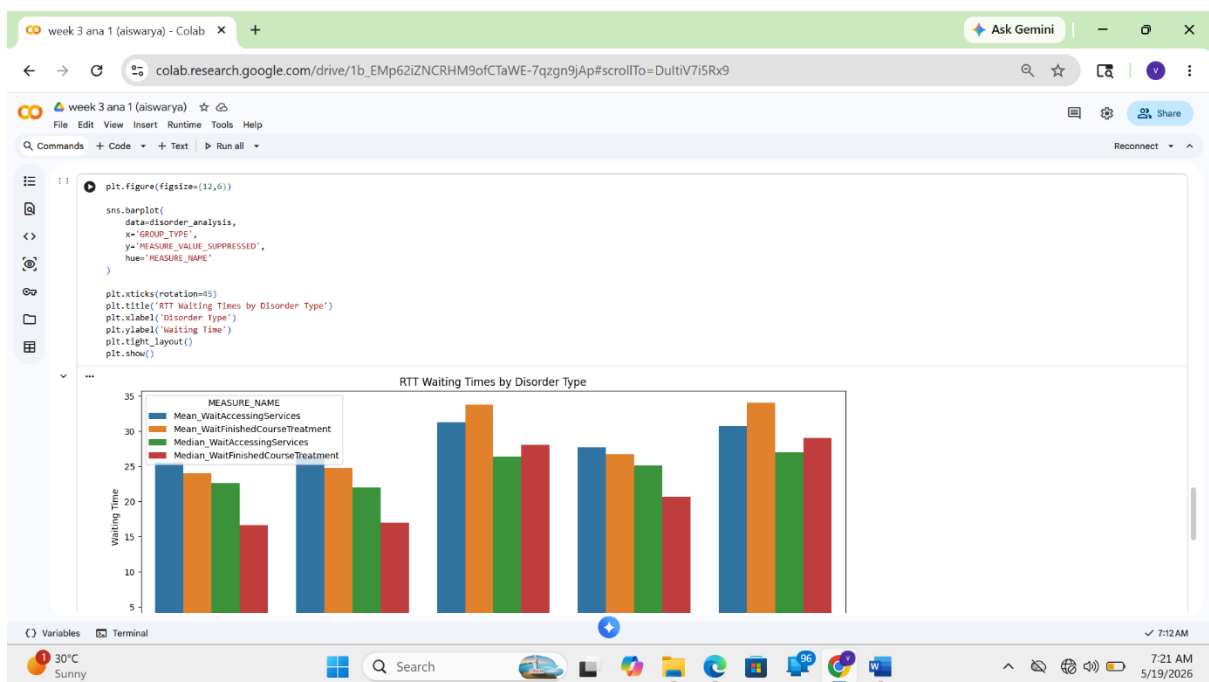
RTT Analysis Chart using Matplotlib

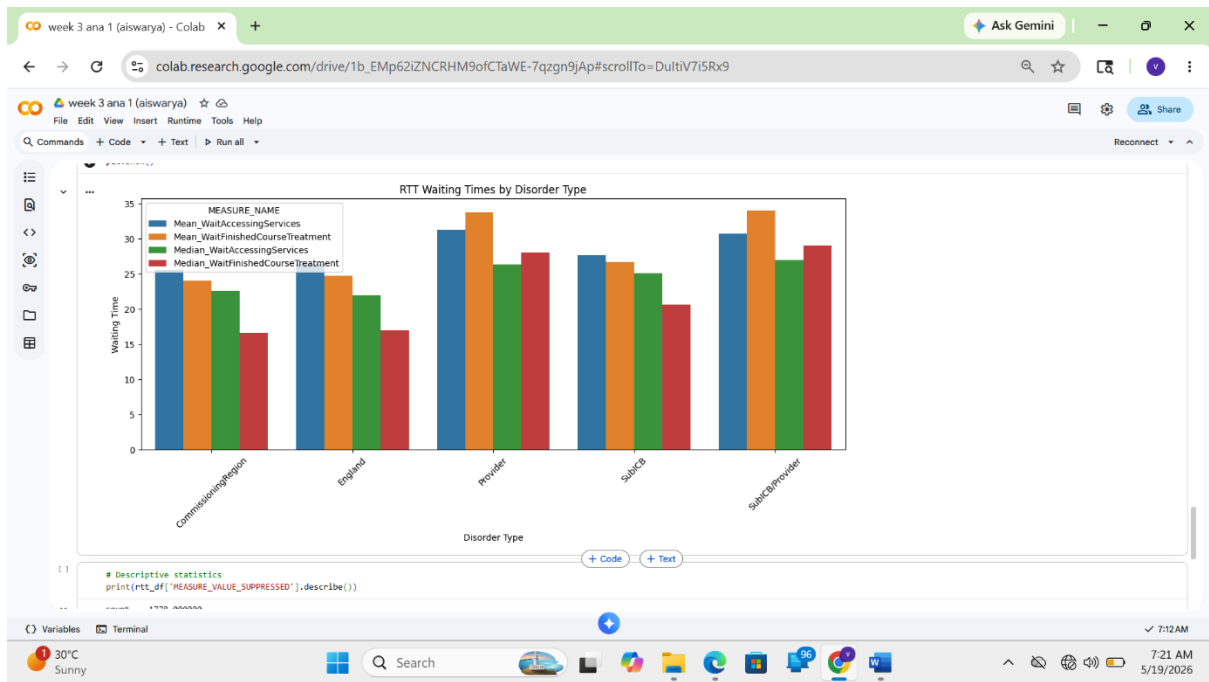


Disorder Type Analysis



Disorder Type Chart





Conclusion

The IAPT RTT analysis provides important insights into mental health treatment accessibility and NHS service performance.

The analysis helps:

- Identify regions with higher waiting times
- Detect disorder groups experiencing treatment delays
- Support healthcare planning and resource allocation